

A SMALL CATALOGUE OF DUST CLOUDS WITHIN 1.25 KPC OF THE SUN

BACHELOR THESIS PRESENTATION

Leon Engel, Advisor: João Alves

April 10, 2025

Universität Wien

TABLE OF CONTENTS

1. Introduction

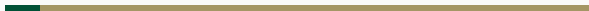
2. Data

3. Methods

4. Results

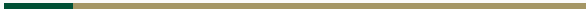
5. Discussion

INTRO – MOTIVATION



- Dust as part of the Interstellar Medium
- Scattering at the Dust Grains
- Reddening of Sources due to Extinction
- Extinction Data inferred from GAIA DR3

DATA



- Concrete Dataset presented by Edenhofer et al. 2024
- Based on $\simeq 54$ Million Stars from (Zhang et al. 2023) within 1.25 kpc.
- A Parsec-Scale 3D Model of the dust distribution

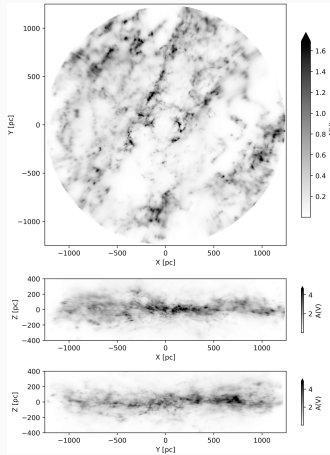
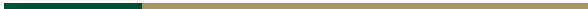


Figure 1: Figure in (Edenhofer et al. 2024, Fig. 5). In Heliocentric Cartesian Coordinates.

METHODS



- Manual selection of cloud regions, in conjunction with Advisor
- Cutoff set to best contain a contiguous Region
- Cylinder around the cloud structure
- Mass calculation in cylindric volume elements, using an adapted method from (O'Neill et al. 2024)
- Error estimation via input parameter variation

Cylinder Parameters:

- Radius r
- Height h
- Center $\vec{t}$
- Angles α, β
- Cut-off density n_C

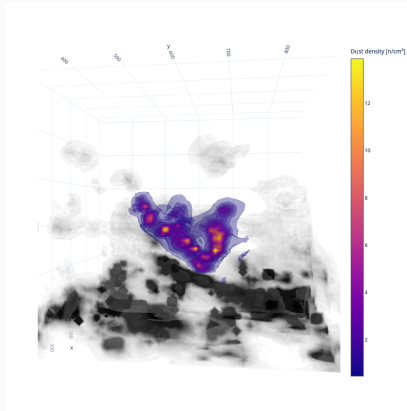


Figure 2: A view of a small part of the Dust distribution model, Cloud 18 of this catalogue in Color.

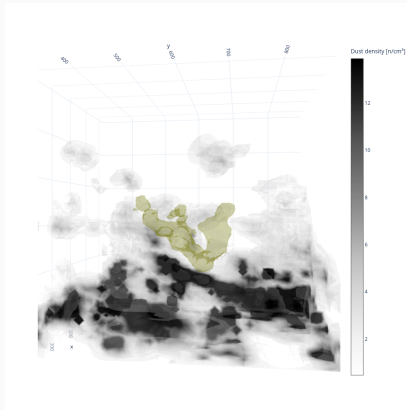


Figure 3: Same as Fig.2, with a density cutoff set at $n_c \simeq 1.4 \text{ cm}^{-3}$.

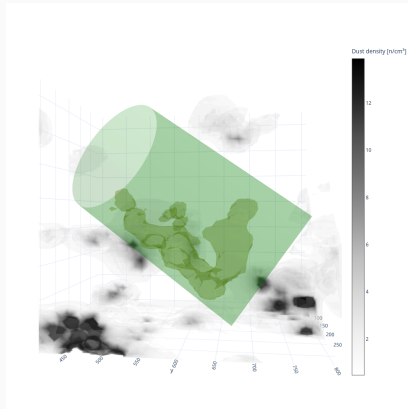


Figure 4: As Fig.3, with the cylinder fit around the cloud.

$$M = (1.57 \pm 0.18) \times 10^4 M_{\odot}$$

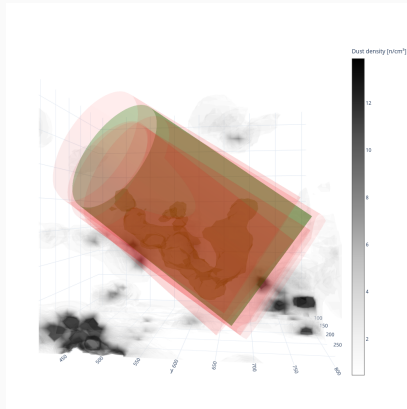


Figure 5: As Fig.4. The red cylinders represent some elements of the process used to estimate the systematic error.

RESULTS



Z-PROJECTION OF THE CATALOGUE

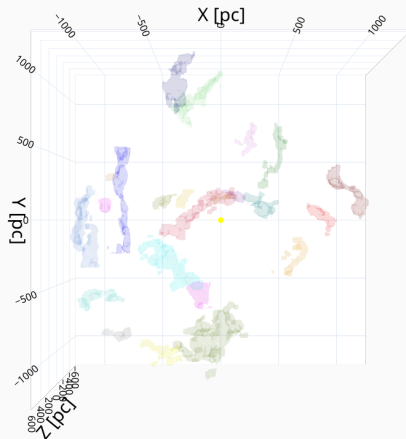


Figure 6: Overview of the clouds from above the galactic plane.

Z-PROJECTION OF THE CATALOGUE

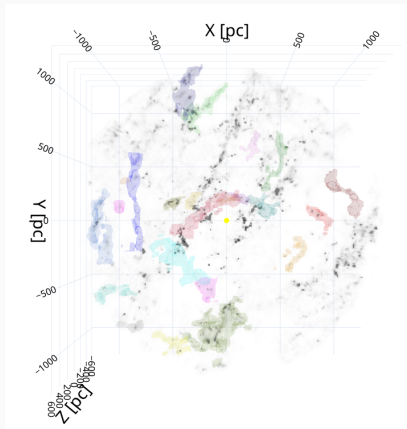


Figure 7: Same as Fig.6, but with an Overlay of the Extinction Map.

SMALLEST CLOUD

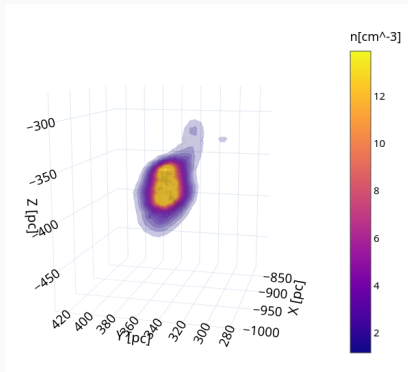


Figure 8: Example of an Element in the Catalogue: Cloud 10,
 $M = (10.77 \pm 0.43) \times 10^3 M_{\odot}$

HIGH-MASS CLOUD ABOVE Z=200

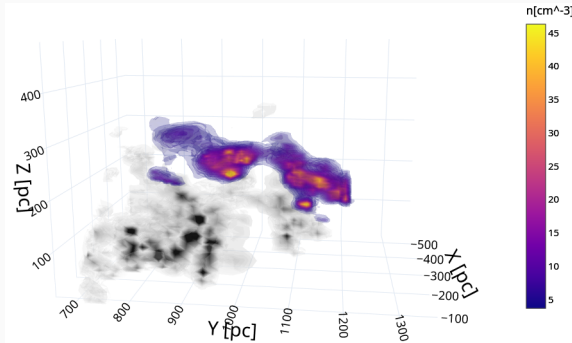


Figure 9: Highest Mass Contiguous Object above $z = 200$ pc,

$$M = (3.07 \pm 0.24) \times 10^5 M_{\odot}$$

CLOUD 1

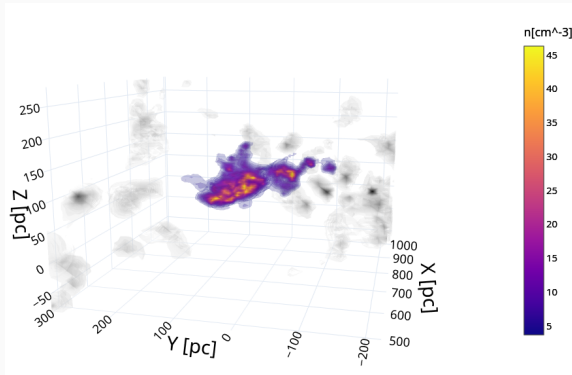


Figure 10: Cloud 1 in this Catalogue, $M = (1.006 \pm 0.039) \times 10^5 M_{\odot}$

DISCUSSION



- Manual Selection of "interesting" Cloud structures
- Very small scales, no analysis of larger structures (Radcliffe Wave, Ceph Spur, etc.)
- Overview of the detail the GAIA data provides Analyses of the ISM

Questions?

REFERENCES

-  Edenhofer, Gordian et al. (May 2024). **“A parsec-scale Galactic 3D dust map out to 1.25 kpc from the Sun”**. In: *Astronomy and Astrophysics* 685. ADS Bibcode: 2024A&A...685A..82E, A82. ISSN: 0004-6361. DOI: 10.1051/0004-6361/202347628. URL: <https://ui.adsabs.harvard.edu/abs/2024A&A...685A..82E> (visited on 02/25/2025).
-  O'Neill, Theo J. et al. (Oct. 2024). **“The Local Bubble Is a Local Chimney: A New Model from 3D Dust Mapping”**. In: *The Astrophysical Journal* 973. Publisher: IOP ADS Bibcode: 2024ApJ...973..136O, p. 136. ISSN: 0004-637X. DOI: 10.3847/1538-4357/ad61de. URL: <https://ui.adsabs.harvard.edu/abs/2024ApJ...973..136O> (visited on 02/25/2025).



Zhang, Xiangyu et al. (Sept. 2023). **“Parameters of 220 million stars from Gaia BP/RP spectra”**. In: *Monthly Notices of the Royal Astronomical Society* 524. Publisher: OUP ADS Bibcode: 2023MNRAS.524.1855Z, pp. 1855–1884. ISSN: 0035-8711. DOI: 10.1093/mnras/stad1941. URL: <https://ui.adsabs.harvard.edu/abs/2023MNRAS.524.1855Z> (visited on 03/05/2025).

- Dataset by Zhang et al. 2023
- BP/RP Spectra of 220 Million Sources
 - ⇒ Extinction from trained Model